

# XRPL Validator — Daily Engineering Report • 2026-04-07

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- 15 commits (xrpl-validator 15)

## Engineering changes

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- **feat(ffl): Stage 2 — shadow state hash from FFI mutations** ( 1a7c435 , xrpl-validator) After FFI applies every tx in a ledger, the accumulated mutations (overlay) are now returned from `apply_ledger_in_order`. The caller can compute a "shadow" state hash by cloning the current FlatHasher, applying the overlay (Created/Modified/Deleted SLEs), and computing the Merkle root — then comparing it to rippled's `account_hash`.
- **fix(ffl): shadow hash must snapshot hasher BEFORE process\_ledger writes — fixes double-apply mismatch** ( a2feb29 , xrpl-validator)
- **fix(ffl): merge protocol-level state changes into shadow overlay — covers singletons + pseudo-tx effects** ( bf46baf , xrpl-validator)
- **fix(ffl): shadow hash uses RPC canonical node\_binary — fixes serialization mismatch** ( 1eaf2a1 , xrpl-validator)
- **fix(ffl): move singleton insertion before shadow hash — shadow overlay now covers all state changes** ( b9d476a , xrpl-validator)
- **feat(ui): shadow hash counter on hero banner** ( 3f51bb2 , xrpl-validator)
- **feat(ui): show shadow hash hex comparison on hero — ours vs network with match indicator** ( 9efb8cc , xrpl-validator)
- **fix(ffl): terPRE\_SEQ recovery — fetch sender AccountRoot at current ledger to break cascade** ( c88e6a8 , xrpl-validator)
- **fix(ffl): skip shadow hash until 5 consecutive state-hash matches — prevents false mismatch during settle** ( 915782c , xrpl-validator)
- **feat: serve static HTML from disk — no rebuild needed for frontend changes** ( 021d687 , xrpl-validator)
- **feat: network topology page (PCB style) + catch-all static route for future pages** ( e4f2140 , xrpl-validator)
- **fix: axum v0.7 path syntax {name} not :name** ( 9aaca2b , xrpl-validator)

## Docs and operations

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- **debug: add overlay size logging to shadow hash mismatch** ( 27261cf , xrpl-validator)
- **docs: test coverage proof — unit tests + live production metrics** ( e9dcf1e , xrpl-validator)
- **debug: add terPRE\_SEQ recovery logging + fallback to ledger-1** ( 6c18f6e , xrpl-validator)

## Appendix — commits

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| Time  | Repo           | Commit  | Subject                                                                                                   |
|-------|----------------|---------|-----------------------------------------------------------------------------------------------------------|
| 06:10 | xrpl-validator | 1a7c435 | feat(ffi): Stage 2 — shadow state hash from FFI mutations                                                 |
| 06:16 | xrpl-validator | a2feb29 | fix(ffi): shadow hash must snapshot hasher BEFORE process_ledger writes — fixes double-apply mismatch     |
| 06:23 | xrpl-validator | bf46baf | fix(ffi): merge protocol-level state changes into shadow overlay — covers singletons + pseudo-tx effects  |
| 06:29 | xrpl-validator | 27261cf | debug: add overlay size logging to shadow hash mismatch                                                   |
| 06:37 | xrpl-validator | 1eaf2a1 | fix(ffi): shadow hash uses RPC canonical node_binary — fixes serialization mismatch                       |
| 06:43 | xrpl-validator | b9d476a | fix(ffi): move singleton insertion before shadow hash — shadow overlay now covers all state changes       |
| 06:56 | xrpl-validator | e9dcf1e | docs: test coverage proof — unit tests + live production metrics                                          |
| 07:52 | xrpl-validator | 3f51bb2 | feat(ui): shadow hash counter on hero banner                                                              |
| 08:00 | xrpl-validator | 9efb8cc | feat(ui): show shadow hash hex comparison on hero — ours vs network with match indicator                  |
| 09:40 | xrpl-validator | c88e6a8 | fix(ffi): terPRE_SEQ recovery — fetch sender AccountRoot at current ledger to break cascade               |
| 10:06 | xrpl-validator | 915782c | fix(ffi): skip shadow hash until 5 consecutive state-hash matches — prevents false mismatch during settle |
| 21:58 | xrpl-validator | 6c18f6e | debug: add terPRE_SEQ recovery logging + fallback to ledger-1                                             |
| 22:30 | xrpl-validator | 021d687 | feat: serve static HTML from disk — no rebuild needed for frontend changes                                |
| 22:47 | xrpl-validator | e4f2140 | feat: network topology page (PCB style) + catch-all static route for future pages                         |
| 22:50 | xrpl-validator | 9aaca2b | fix: axum v0.7 path syntax {name} not :name                                                               |